

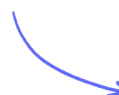
Non-Calculator Questions

Expanding & Factorising Brackets

Expanding & Simplifying Single Brackets / Expanding Double Brackets / Expanding Triple Brackets / Factorising Out Terms / Factorising by Grouping / Factorising Simple Quadratics / Factorising Harder Quadratics / Difference of Two Squares / Deciding the Factorisation Method

Easy (9 questions)	/9
Medium (9 questions)	/21
Hard (12 questions)	/27
Very Hard (5 questions)	/13
Total Marks	/70

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Easy Questions

1 Factorise $5p + pt$.

(1 mark)

2 Expand.

$$a(a^3 + 3)$$

(1 mark)

3 Factorise.

$$12x + 15$$

(1 mark)

4 Factorise $5y - 6py$.

(1 mark)

5 Factorise $2x^2 - x$.

(1 mark)

6 Factorise $7k^2 - 15k$.

(1 mark)

7 Factorise.

$$y - 2y^2$$

(1 mark)

8 Expand.

$$7(x - 8)$$

(1 mark)

9 Factorise.

$$w + w^3$$

(1 mark)

Medium Questions

- 1 Factorise completely $21a^2 + 28ab$.

(2 marks)

- 2 Expand and simplify.

$$(m - 3)(m + 2)$$

(2 marks)

- 3 Expand and simplify $(x + 3)(x + 5)$.

(2 marks)

- 4 Expand and simplify.

$$(3x - 7)(2x + 9)$$

(2 marks)

- 5 Expand.

$$2x(3 - x^2)$$

(2 marks)

6 Expand and simplify.

$$6(2y - 3) - 5(y + 1)$$

(2 marks)

7 Expand the brackets and simplify.

$$(2p + 3)(3p - 2)$$

(3 marks)

8 Factorise completely $15k^2m - 20m^4$.

(2 marks)

9 Expand and simplify.

$$(x - 5)(2x - 1)(x + 3) + x(x + 4)$$

(4 marks)

Hard Questions

- 1 Expand and simplify.

$$x(x + 2)(x - 3)$$

(3 marks)

- 2 Factorise $6x^2 + 7x - 20$.

(2 marks)

- 3 Factorise $3x + 8y - 6ax - 16ay$.

(2 marks)

4 $x^2 - 12x + a = (x + b)^2$

Find the value of a and the value of b .

$$a = \dots\dots\dots$$

$$b = \dots\dots\dots$$

(2 marks)

- 5 Factorise.

$$xy - 2x + 3y - 6$$

(2 marks)

6 Expand and simplify $(x + 1)(x + 2) + 2x(x - 3)$.

(3 marks)

7 (a) Factorise $p^2 - q^2$.

(1 mark)

(b) $p^2 - q^2 = 7$ and $p - q = 2$.

Find the value of $p + q$.

(2 marks)

8 Complete this statement with an expression in terms of m .

$$18m^3 + 9m^2 + 14m + 7 = (9m^2 + 7)(\dots\dots\dots)$$

(2 marks)

9 Factorise.

$$xy + 5y + 2x + 10$$

(2 marks)

10 Factorise completely.

$$px + py - x - y$$

(2 marks)

11 Factorise completely.

$$xy + 2y + 3x + 6$$

(2 marks)

12 Factorise completely.

$$2a + 4b - ax - 2bx$$

(2 marks)

Very Hard Questions

1 Factorise completely $20x^2 - 45y^2$.

(3 marks)

2 Expand and simplify $(x + 3)(x - 5)(3x - 1)$.

(3 marks)

3 Factorise.

$$18y - 3ay + 12x - 2ax$$

(2 marks)

4 Factorise.

$$3x^2 - 48y^2$$

(3 marks)

5 Factorise $12(m + p) + 8(m + p)^2$.

(2 marks)